

ECON 351 Exam Prep

24-Question Practice Set

Sources: Homework 1 & 2, Sample Questions Ch.4 & Ch.9, Mock Midterm 1.

Topic counts match professor breakdown (12 Ch.4 + 12 Ch.9).

Labor-leisure Q3-Q4 are course-style (not in uploads).

Study Plan

Chapter 4 (12)

- 2 Consumer Problems
- 2 Labor-Leisure
- 2 Intertemporal Consumption
- 3 Elasticity / Types of Goods
- 1 Budget Line
- 2 Assumptions / Preferences

Chapter 9 (12)

- 5 Insurance Problems
- 4 Risk Attitudes
- 2 Certainty Equivalent & Risk Premium
- 1 Diversification

Chapter 4 - Practice Questions

Q1 - Consumer Problem #1

Consumer Problems | Homework 1 Ch.4 (Ex 3.1-3.5)

Manuel: $U(q_1, q_2) = 3q_1^{0.6} q_2^{0.4}$. Prices $p_1=5$, $p_2=10$, income=\$100. Find q_1^* , q_2^* .

Note: Full consumer theory problem.

Q2 - Consumer Problem #2

Consumer Problems | Sample Q Ch.4 (Q26)

$PA=\$2$, $PB=\$3$, $I=\$120$, $U(A,B)=2A^2B^3$. Write budget constraint, find optimal bundle, slopes at optimum.

Q3 - Labor-Leisure #1

Labor-Leisure | Course-style (not in uploads) (LL-1)

Maria has 16 hours/day for leisure (l) and work. Wage $w=\$20/\text{hr}$, non-labor income $V=\$80$. Utility $U(c,l)=c^{0.5} l^{0.5}$ where $c = w(16-l)+V$. Find optimal l^* and labor hours L^* .

Note: Standard labor-leisure Cobb-Douglas.

Q4 - Labor-Leisure #2

Labor-Leisure | Course-style (not in uploads) (LL-2)

Same setup as LL-1 but wage rises to $w=\$30$. Find new l^* , L^* , c^* . Does Maria work more or less?

Note: Compare to LL-1.

Q5 - Intertemporal #1

Intertemporal Consumption | Sample Q Ch.4 (Q41)

$I_1=10$, $I_2=12$, $r=0.5$, $p_1=p_2=1$, $u(q_1, q_2)=q_1^*q_2$. Find q_1^* , q_2^* . Saving or borrowing?

Q6 - Intertemporal #2

Intertemporal Consumption | Mock Midterm 1 (Q25)

Using Q24 setup, optimal q_1 in period 1?

Q7 - Elasticity #1

Elasticity / Types of Goods | Homework 1 Ch.4 (Ex 4.1-4.2)

Demand $q=100-5p$. Find Q at $p=5$ and $p=10$. Compute price elasticity at $p=5$.

Q8 - Elasticity #2

ECON 351 - Exam Prep (24-Question Practice Set)

Elasticity / Types of Goods | Sample Q Ch.4 (Q39)

QA = 784 ? 2PA ? 3I ? 4PO. At PA=2, I=100, PO=20: price, income, and cross-price elasticities.

Q9 - Elasticity #3

Elasticity / Types of Goods | Sample Q Ch.4 (Q38)

Apples & bananas complements; apples inferior. Which demand function for apples?

Note: Negative on I (inferior), positive on PB (complement).

Q10 - Budget Line

Budget Line | Sample Q Ch.4 (Q10)

Apples (horizontal), oranges (vertical). Price of apples rises \$3->\$6; price of oranges rises \$2->\$4. Budget line:

Note: Slope unchanged (?PA/PO); income effectively halved.

Q11 - Preferences #1

Assumptions / Preferences | Homework 1 Ch.4 (Ex 2.2)

Jackie goes to the movies. Movies: 'Avatar' and 'The Avengers.' She cannot decide which one to see. Does this violate any assumption on preferences?

Q12 - Preferences #2

Assumptions / Preferences | Sample Q Ch.4 (Q8)

Serena's IC is a straight line with slope $7/5$ (X on horizontal). Which utility function represents her preferences? Options include $u = X/5 + Y/7$.

Note: IC slope = MRS = $(1/5)/(1/7) = 7/5$.

Chapter 9 - Practice Questions

Q13 - Insurance #1

Insurance | Homework 2 Ch.9 (Ex 3a)

Tom: house \$846,400, 25% chance \$174,000 damage, $u = \sqrt{x}$. Max premium P for full insurance?

Q14 - Insurance #2

Insurance | Homework 2 Ch.9 (Ex 3b)

Same Tom setup. Fair (actuarially fair) insurance premium?

Q15 - Insurance #3

Insurance | Sample Q Ch.9 (Q18)

Sherman: Ferrari \$230,900 + \$50,000 cash. 25% chance \$158,400 damage. $u = \sqrt{x}$. Max P and RP ?

Q16 - Insurance #4

Insurance | Mock Midterm 1 (Q27)

\$1,000 cash. 10% sick, doctor \$500. Premium \$100, copay \$275 if sick. $u = \sqrt{x}$. $E[u]$ with insurance?

Q17 - Insurance #5

Insurance | Mock Midterm 1 (Q25-26)

Car \$10,000, 75% crash \$3,600 damage. $u = \sqrt{x}$. Max P and RP ?

Q18 - Risk Attitudes #1

Risk Attitudes | Homework 2 Ch.9 (Ex 2)

Mike's $MU(I)$ graph given. Bet: pay \$20 on dice 1-2, receive \$10 on 3-6. Fair bet? Accept?

Q19 - Risk Attitudes #2

Risk Attitudes | Sample Q Ch.9 (Q6)

Anna $U(I) = I$, Elsa $U(I) = I^{1.5}$. Anna is ____ and Elsa is ____.

Note: $a > 1 \rightarrow$ risk loving; $a = 1 \rightarrow$ neutral; $a < 1 \rightarrow$ averse.

Q20 - Risk Attitudes #3

Risk Attitudes | Sample Q Ch.9 (Q8)

Graphs of $u(I)$ for Marta and Hannah. Classify both.

Note: Concave vs convex utility.

Q21 - Risk Attitudes #4

Risk Attitudes | Mock Midterm 1 (Q28)

Firmino bet: Reds win $\rightarrow$ friend pays \$1; lose $\rightarrow$ Firmino pays \$3. Mohamed $U=I^{0.7}$, Sadio $U=I^{2.3}$.

Q22 - Certainty Eq. & RP #1

Certainty Equivalent & Risk Premium | Homework 2 Ch.9 (Ex 1)

Investment: 80% $\rightarrow$ \$200, 20% $\rightarrow$ \$50. $u(x)=\sqrt{x}$. Find $E[x]$, $E[u]$, certainty equivalent, risk premium.

Q23 - Certainty Eq. & RP #2

Certainty Equivalent & Risk Premium | Sample Q Ch.9 (Q12)

80% $\rightarrow$ \$144, 20% $\rightarrow$ \$81, $u(x)=\sqrt{x}$. Find $E[u]$, $E[x]$, CE, RP.

Q24 - Diversification

Diversification | Sample Q Ch.9 (Q22)

Feng: companies A&B (agriculture, positively correlated). Nan: C&D (airline/oil, negatively correlated). Who has riskier portfolio?

Formula Sheet - Chapter 4

Budget constraint: $p_1q_1 + p_2q_2 = I$ (Slope = $-p_1/p_2$)

Optimal (interior): $MRS = MU_1/MU_2 = p_1/p_2$ (Tangency condition)

Cobb-Douglas $U = q_1^a q_2^b$: $q_1^* = a/(a+b) * I/p_1$ ($q_2^* = b/(a+b) * I/p_2$)

Perfect substitutes: Buy only good with higher MU/p (Corner solution)

Perfect complements: Consume at fixed ratio; kink on IC ($q_1/q_2 = \text{constant}$)

MRS (Cobb-Douglas): $MRS = (a/b)(q_2/q_1)$

Labor-Leisure: $c = wL + V$, $L = T - l$ (T = total time; V = non-labor income)

Labor-Leisure optimum: $MU_l / MU_c = w$ (Cobb-Douglas: $l^* = (wT+V)/(2w)$ if $a=b=0.5$)

Intertemporal BC: $c_1 + c_2/(1+r) = I_1 + I_2/(1+r)$ (Discount future consumption)

Intertemporal optimum: $MU_1/MU_2 = (1+r)$ (Future price = $1+r$)

Present value: $PV = FV/(1+r)$ (Compare contracts by PV)

Price elasticity: $E_p = (P/Q)(\partial Q/\partial P)$ (Linear: $E_p = (\partial Q/\partial P)*(P/Q)$)

Income elasticity: $E_I = (I/Q)(\partial Q/\partial I)$ ($E_I > 0$ normal; $E_I < 0$ inferior)

Cross-price elasticity: $E_{xy} = (P_y/Q_x)(\partial Q_x/\partial P_y)$ (+ substitutes; - complements; 0 independent)

Formula Sheet - Chapter 9

Expected value: $E[x] = \sum p_i x_i$ (Probability-weighted average)

Expected utility: $E[u] = \sum p_i u(x_i)$ (EU theory)

Risk attitudes: $u''(x) < 0 \rightarrow$ risk averse ($u''=0$ neutral; $u''>0$ loving)

MU test: Decreasing MU(l) $\rightarrow$ risk averse (Constant MU $\rightarrow$ neutral; Increasing $\rightarrow$ loving)

Certainty equivalent (CE): $u(CE) = E[u(x)]$ (Sure \$ with same EU as gamble)

Risk premium (RP): $RP = E[x] - CE$ (Extra $E[x]$ to accept risk)

Fair bet: $E[\text{payoff}] = 0$ (Risk averse rejects; lover accepts)

Fair insurance premium: $P_{\text{fair}} = E[\text{loss}] = \sum p_i \cdot \text{loss}_i$ (Actuarially fair)

Max insurance premium: $E[u(\text{with ins})] = E[u(\text{no ins})]$ (Solve for P)

Insurance risk premium: $RP_{\text{ins}} = P_{\text{max}} - P_{\text{fair}}$

Partial insurance EU: $E[u] = \sum p_i u(\text{income in state } i)$ (Include premium all states)

Diversification: Negatively correlated assets $\rightarrow$ portfolio variance (Don't put eggs in one basket)

Answer Key

Q1 - Consumer Problem #1

MRS = $(0.6/0.4)(q_2/q_1) = p_1/p_2 \rightarrow q_2 = (2/3)q_1$. Budget: $5q_1 + 10q_2 = 100 \rightarrow q_1^* = 12, q_2^* = 8$.

Q2 - Consumer Problem #2

BC: $2A + 3B = 120$, slope = $-2/3$. Optimum: $A^* = 24, B^* = 24$. MRS at X = $3B/(2A) = 3/2$.

Q3 - Labor-Leisure #1

MRS: $I/c = w \rightarrow I = 20(16?I) + 80 = 400?20I \rightarrow I^* = 400/21?19.05 \rightarrow$ capped at 16? Recheck: $I = wL + V/w$ with $L = 16?I \rightarrow I^* = (w^*16 + V)/(2w) = (320 + 80)/40 = 10$. $I^* = 10, L^* = 6, c^* = 200$.

Q4 - Labor-Leisure #2

$I^* = (30^*16 + 80)/(2^*30) = (480 + 80)/60 = 560/60 ? 9.33$. $L^* ? 6.67$. She works MORE (substitution effect dominates for Cobb-Douglas).

Q5 - Intertemporal #1

PV budget: $q_1 + q_2/1.5 = 10 + 12/1.5 = 18$. Optimum $q_1 = q_2$ (Cobb-Douglas $a = 0.5$). $q_1^* = 9, q_2^* = 13.5$. Consumes more in period 2 $\rightarrow$ saving.

Q6 - Intertemporal #2

(D) $q_1 = 80$

Q7 - Elasticity #1

$Q(5) = 75, Q(10) = 50$. $E_p = (P/Q)(dQ/dP) = (5/75)(?5) = ?1/3$.

Q8 - Elasticity #2

$Q_A = 400$. $E_p = ?1, E_I = ?0.75$ (inferior), $E_{XO} = ?0.20$ (complements).

Q9 - Elasticity #3

(C) $Q_A = 110 ? 2P_A ? 4I + 2P_B$

Q10 - Budget Line

(C) shift inward (parallel shift).

Q11 - Preferences #1

Yes - violates completeness (consumer cannot rank bundles).

Q12 - Preferences #2

(D) $u(X, Y) = X/5 + Y/7$

Q13 - Insurance #1

$P_{\max} ? \$45,375$ (EU with insurance = EU without at $P = 45,375$).

Q14 - Insurance #2

$E[\text{loss}] = 0.25 \times \$174,000 = \$43,500$.

Q15 - Insurance #3

$P_{\max} ? \$45,675$. Fair = \$39,600. RP ? \$6,075.

Q16 - Insurance #4

(B) 30.7

Q17 - Insurance #5

$P_{\max} ? \$2,775$ (D). $RP = \$25$ (E).

Q18 - Risk Attitudes #1

Fair bet ($E[\text{win}] = 0$). Risk averse $\rightarrow$ reject. Does not depend on wealth if can pay \$20.

Q19 - Risk Attitudes #2

(A) risk neutral, risk loving.

Q20 - Risk Attitudes #3

(C) Marta risk averse; Hannah risk loving.

Q21 - Risk Attitudes #4

(D) Mohamed rejects; Sadio accepts.

Q22 - Certainty Eq. & RP #1

$E[x] = 170$. $E[u] = 0.8\sqrt{200} + 0.2\sqrt{50} ? 12.66$. CE: $\sqrt{CE} = 12.66 \rightarrow CE ? 160.4$. $RP = 170 ? 160.4 ? 9.6$.

Q23 - Certainty Eq. & RP #2

$E[x] = 131.4$. CE ? 129.96. RP ? 1.44.

Q24 - Diversification

Feng - positively correlated assets don't diversify; Nan's portfolio has lower variance.